

Universal Basic Income and Social Welfare Under AI-Driven Labor Market Scenarios

A U.S. Microsimulation with Evidence-Calibrated AI and UBI Assumptions

UBI-AI Welfare Project

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Executive summary. This policy brief estimates how Universal Basic Income (UBI) affects social welfare under three AI labor-market scenarios for the United States. Using a transparent household microsimulation calibrated to IMF, OpenAI/UPenn, and RCT evidence, we find that **UBI welfare gains are largest when AI threatens job security** (misaligned automation). At \$1,000/month deficit-financed UBI under that scenario, the simulated poverty rate falls to **0.9%** with a **+0.00%** change in aggregate social welfare. Gains concentrate in lower income deciles, but net fiscal costs rise sharply at higher benefit levels.

Problem and Policy Question

Artificial intelligence may complement or substitute for human labor. Universal Basic Income has been proposed as an earnings buffer during technological transition. **This report asks:** *How much social welfare does UBI add, and how does that depend on whether AI augments or displaces workers?*

We evaluate three baseline paths:

1. **Historical innovation** – disruption with reallocation and productivity (web / industrial-revolution analogy)
2. **Aligned augmentation** – AI complements workers; low displacement
3. **Misaligned automation** – AI substitutes workers; high displacement and earnings volatility

UBI is modeled as \$250–\$1,500 per adult per month with deficit, flat-tax, and progressive-tax funding.

Evidence Base

AI and employment

Source	Key estimate
IMF (International Monetary Fund 2024)	~40% global employment exposed; ~60% in advanced economies
Eloundou et al. (Eloundou et al. 2023)	80% of U.S. workers have $\geq 10\%$ of tasks LLM-exposed
Goldman Sachs (Goldman Sachs 2023)	~6-7% worker displacement in broad adoption case
Budget Lab (Yale Budget Lab 2025)	Limited aggregate displacement through 2024-25

UBI and cash transfers

Study	Finding
OpenResearch (Vivalt et al. 2024)	\$1,000/mo: -2 to -4 pp labor participation; higher income net of transfers
Stockton SEED (West and Castro 2023)	\$500/mo: improved stability; no strong labor harm
Alaska PFD (Jones and Marinescu 2022)	No aggregate employment effect; +1.8 pp part-time work

Data and Scenarios

Table 1: AI scenario calibration (see scenarios.yml for sources).

Scenario	Displacement rate	Augmentation share	Productivity gain	Shock intensity
historical_innovation Historical Innovation (Web / Industrial Revolution)	0.065	0.50	0.15	0.45
aligned_augmentation Aligned Augmentation (Complementary AI)	0.020	0.63	0.10	0.25
misaligned_automation Misaligned Automation (Substituting AI)	0.120	0.20	0.05	0.85

Our baseline uses **5000** synthetic U.S. households mapped to occupation-group AI exposure (Eloundou et al. 2023; Felten et al. 2021). CPS ASEC microdata can replace this when Census API access is available.

Methods

For each household we: (1) apply an AI income shock by occupation exposure; (2) add UBI transfers; (3) scale labor supply using OpenResearch elasticities (Vivalt et al. 2024); (4) apply financing rules; (5) compute CRRA utility ($\rho = 2$) and consumption-equivalent variation (Colombino 2023).

Quasi-experimental templates (Economics 50 / (Chetty and Bruich 2019)): DiD/event study around ChatGPT (Nov 2022), Alaska-style synthetic control (Jones and Marinescu 2022), and ML prediction of household-level welfare gains. These are **illustrative templates**, not causal claims.

Results

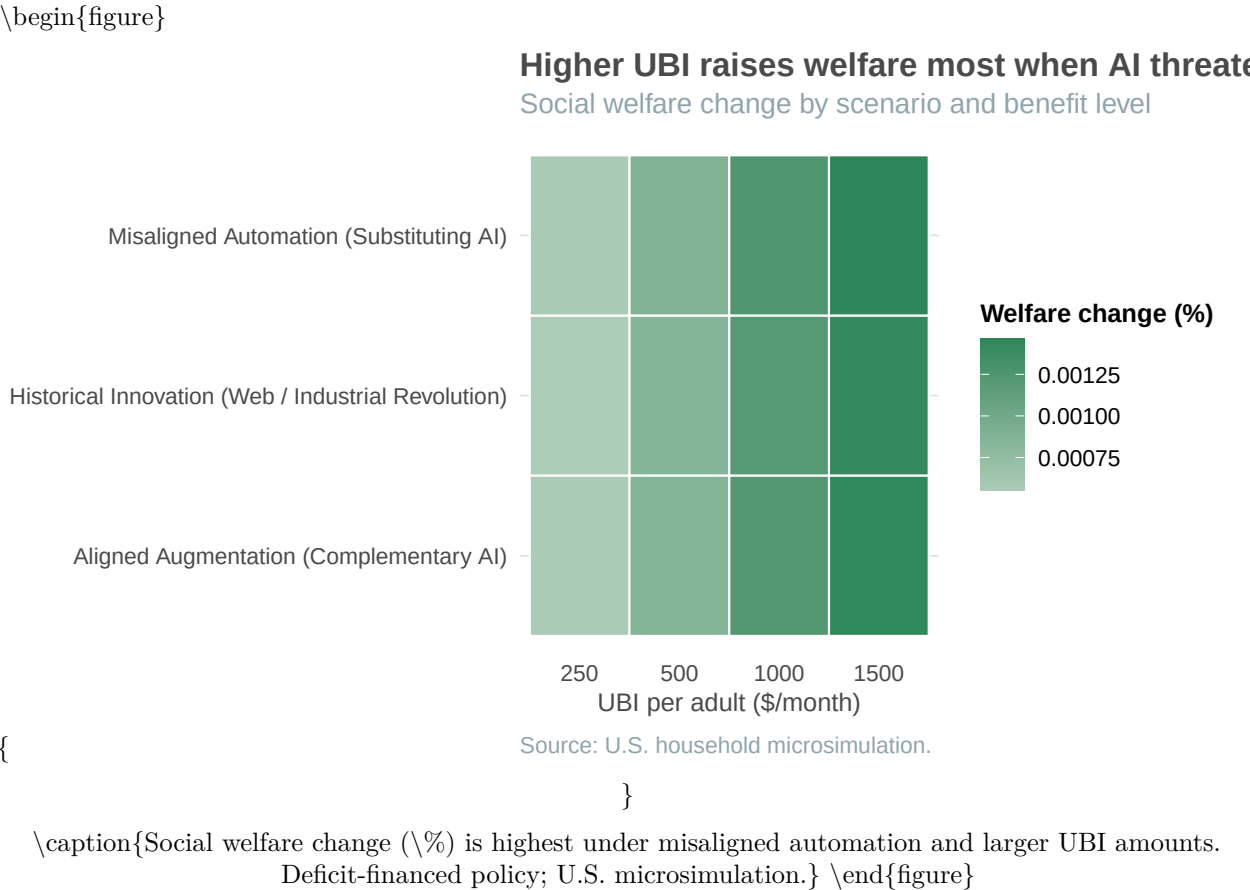
UBI reduces poverty across all AI scenarios

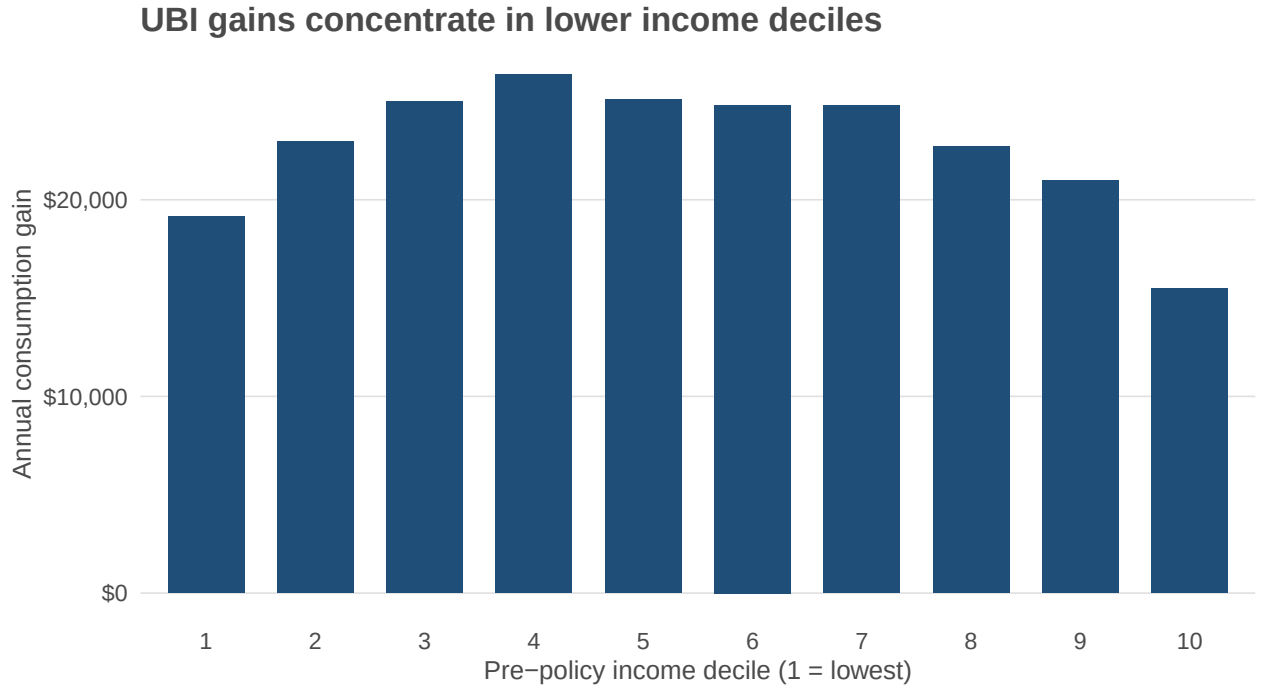
Table 2: Deficit-financed UBI: selected outcomes.

AI scenario	UBI /mo Poverty)				
Historical Innovation (Web / Industrial Revolution)	250	10.9	0.440	0	0.04
Historical Innovation (Web / Industrial Revolution)	500	5.2	0.412	0	0.08
Historical Innovation (Web / Industrial Revolution)	1000	0.9	0.368	0	0.16

AI scenario	UBI <i>/mo Poverty)</i>				
Historical Innovation (Web / Industrial Revolution)	1500	0.1	0.336	0	0.24
Aligned Augmentation (Complementary AI)	250	11.2	0.440	0	0.04
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Aligned Augmentation (Complementary AI)	1000	0.9	0.368	0	0.16
Aligned Augmentation (Complementary AI)	1500	0.1	0.335	0	0.24
Misaligned Automation (Substituting AI)	250	11.4	0.440	0	0.04
Misaligned Automation (Substituting AI)	500	5.3	0.411	0	0.08
Misaligned Automation (Substituting AI)	1000	0.9	0.367	0	0.16
Misaligned Automation (Substituting AI)	1500	0.1	0.334	0	0.24

Welfare gains rise with UBI generosity and AI disruption





{ Misaligned automation; \$1,000/mo UBI. }

\caption{Under \\$1,000/month UBI and misaligned automation, lower deciles see the largest consumption gains.} \end{figure}

Continuous AI shock sensitivity

Quasi-experimental templates

Difference-in-differences: DiD treated coefficient = -8×10^{-4} (template panel; no clear break detected).

Synthetic control: Alaska template ATT = **-0.01**, consistent with no aggregate employment decline (Jones and Marinescu 2022).

Policy Implications

- **If AI augments workers:** UBI still reduces poverty, but the welfare case for large transfers is weaker because baseline disruption is limited.
- **If AI displaces workers:** UBI provides the strongest buffer, especially for high-exposure, lower-income households, but fiscal costs are highest.
- **Labor supply matters:** Scaling OpenResearch elasticities (Vivalt et al. 2024), earned income falls modestly; progressive financing further reduces net consumption gains.

Limitations

- Synthetic household data (CPS ASEC pipeline available via `cpsR`)
- AI parameters calibrated to published ranges, not dynamically forecast
- Quasi-experimental modules are templates requiring validated data
- CRRA $\rho = 2$ is illustrative; robustness to $\rho \in [1, 3]$ recommended

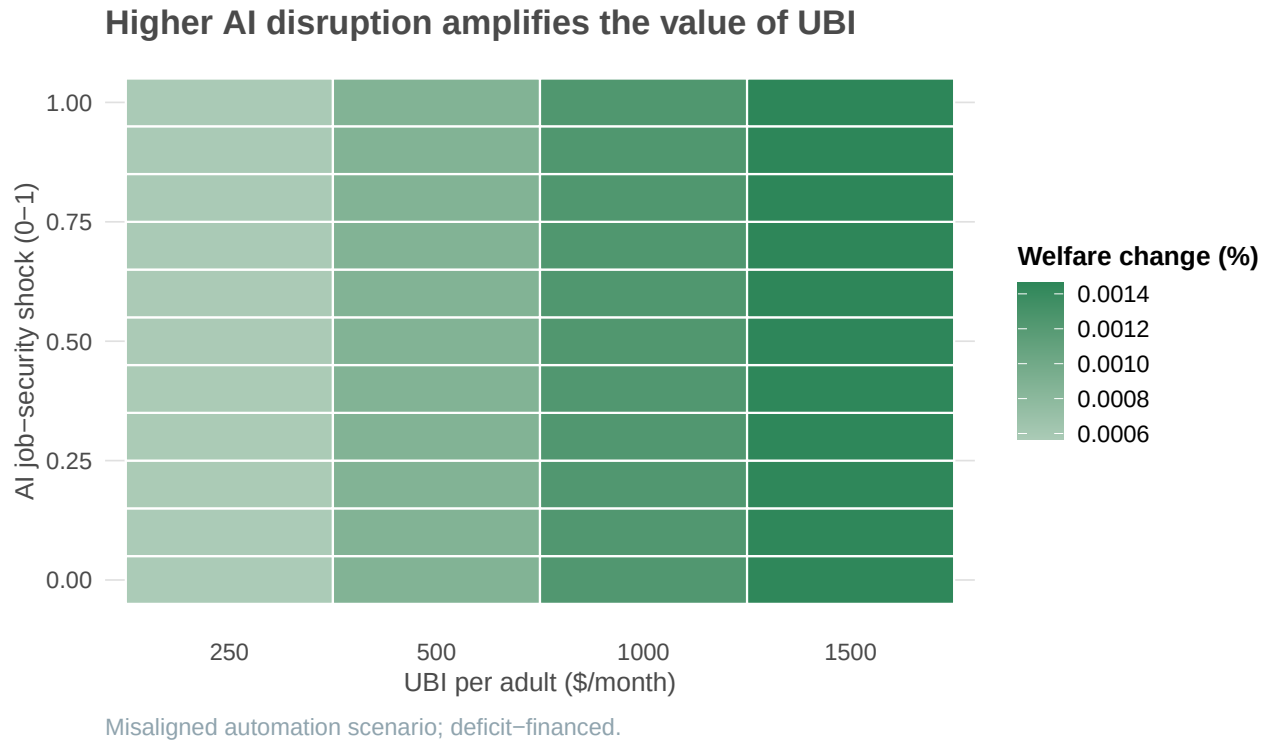


Figure 1: Welfare gains increase with both AI job-security shock intensity and UBI generosity.

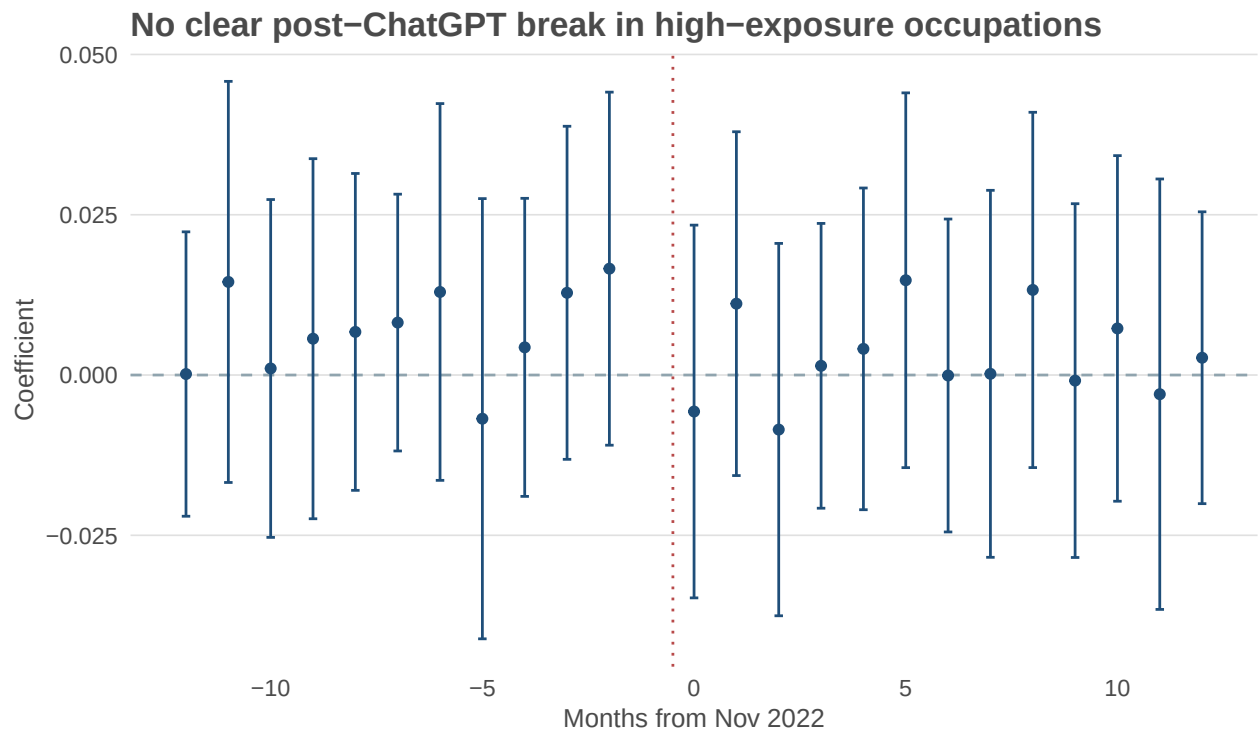


Figure 2: Event-study template: high vs. low AI exposure after ChatGPT release.

Conclusion

UBI’s social welfare value depends critically on how AI affects job security. Under misaligned automation, unconditional transfers provide the strongest distributional buffer. Policy design must weigh poverty reduction and welfare gains against fiscal cost and documented labor supply responses.

Interactive dashboard: <http://127.0.0.1:3838> (`run shiny::runApp(".")`).

References

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